

Project Report: Microsoft Fabric Lakehouse Analytics - Power BI Analyst

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Family:	Power BI Analyst
Level:	Advanced
Chapters:	6
Source:	CodeQuest project specification (official docs + industry patterns)

Summary

Build Fabric lakehouse: ingest raw data, transform in notebooks, semantic model, and executive dashboard.

Chapters

Track: Project-Intro

Chapter 1: Project Overview [Advanced]

Chapter focus: Project Overview *(Level: Advanced)**

Microsoft Fabric Lakehouse Analytics is a advanced-level end-to-end project for Power BI Analyst. Duration: 5-6 weeks. Build Fabric lakehouse: ingest raw data, transform in notebooks, semantic model, and executive dashboard. Tools: Microsoft Fabric, Power BI, OneLake.**

Code Reference:

Objective: Create Fabric workspace and lakehouse
Objective: Ingest CSV/Parquet with pipelines
Objective: Build semantic model in Direct Lake mode
Objective: Executive dashboard with drill-through

What it shows: Lists measurable outcomes before you start building.

Topic guide

What it actually is

A project overview defines scope, tools, and success criteria.

When to use it

At project kickoff.

Where to use it

Power BI Analyst portfolios and CodeQuest project submissions.

Real use example

A candidate lists this project on a Power BI Analyst resume with a demo link.

Track: Project-Phase

Chapter 2: Phase: Lakehouse [Advanced]

Chapter focus: Phase: Lakehouse *(Level: Advanced)**

OneLake setup. Complete these tasks: Shortcuts; Tables. Deliverable for this phase: Lakehouse schema.

Code Reference:

- [] Shortcuts
- [] Tables

What it shows: Checklist format helps track phase completion.

Topic guide**What it actually is**

Phase Lakehouse is one step in the full project lifecycle.

When to use it

During week 1 of the project timeline.

Where to use it

Power BI Analyst agile sprints and coursework milestones.

Real use example

Team demo shows the Lakehouse schema to the teacher.

Chapter 3: Phase: Pipeline [Advanced]

Chapter focus: Phase: Pipeline *(Level: Advanced)**

Data Factory pipeline. Complete these tasks: Schedule refresh; Error handling. Deliverable for this phase: Pipeline run history.

Code Reference:

- [] Schedule refresh
- [] Error handling

What it shows: Checklist format helps track phase completion.

Topic guide

What it actually is

Phase Pipeline is one step in the full project lifecycle.

When to use it

During week 2 of the project timeline.

Where to use it

Power BI Analyst agile sprints and coursework milestones.

Real use example

Team demo shows the Pipeline run history to the teacher.

Chapter 4: Phase: Semantic [Advanced]

Chapter focus: Phase: Semantic *(Level: Advanced)**

Direct Lake model. Complete these tasks: Relationships; Aggregations. Deliverable for this phase: Semantic model.

Code Reference:

- [] Relationships
- [] Aggregations

What it shows: Checklist format helps track phase completion.

Topic guide**What it actually is**

Phase Semantic is one step in the full project lifecycle.

When to use it

During week 3 of the project timeline.

Where to use it

Power BI Analyst agile sprints and coursework milestones.

Real use example

Team demo shows the Semantic model to the teacher.

Chapter 5: Phase: Dashboard [Advanced]

Chapter focus: Phase: Dashboard *(Level: Advanced)**

Executive landing page. Complete these tasks: KPI cards; Drill-through. Deliverable for this phase: Published Fabric report.

Code Reference:

- [] KPI cards
- [] Drill-through

What it shows: Checklist format helps track phase completion.

Topic guide

What it actually is

Phase Dashboard is one step in the full project lifecycle.

When to use it

During week 4 of the project timeline.

Where to use it

Power BI Analyst agile sprints and coursework milestones.

Real use example

Team demo shows the Published Fabric report to the teacher.

Track: Project-Deliverables

Chapter 6: Final Deliverables and Review [Advanced]

Chapter focus: Final Deliverables and Review** *(Level: Advanced)

Submit all final deliverables: Fabric workspace; Pipeline definition; Published report; Architecture diagram. Skills demonstrated: Fabric, Lakehouse, Pipelines. Prepare a 5-minute walkthrough video and README for CodeQuest teacher marking.

Code Reference:

```
Deliverable: Fabric workspace
Deliverable: Pipeline definition
Deliverable: Published report
Deliverable: Architecture diagram
```

What it shows: Final checklist ensures nothing is missing at submission.

Topic guide

What it actually is

Deliverable review confirms end-to-end project completion.

When to use it

Before deadline and teacher review.

Where to use it

CodeQuest project gallery and interviews.

Real use example

A student submits Fabric workspace and passes the rubric.